



Evo2 Variant Intelligence

☰ Can show publicly	Yes
🏠 Category	Biotechnology / Genomics
🌐 Country	Morocco
🔗 Industry	Genomics, biotech, health AI, research intelligence
☰ Main features	- - -
🔗 Project link	https://evo2-variant-intelligence.vercel.app/
☰ Project type	Client Work / Internal Product
☰ Proof available	Public link: Screenshots: Demo: Testimonials: Metrics:
☰ Status	Delivered
☰ Technical stack	- - -

I'll conduct a comprehensive audit of both repositories. Let me start by examining their structure and contents. Now let me explore the repository structure and key files: Based on my comprehensive analysis of both repositories, here is the complete audit report:

Project Repository Report

1. Project identity

- **Project name:** Evo2 Variant Pathogenicity Prediction Platform
 - **Repository names:** `khaireddine-arbouch/evo2-API` (backend) and `khaireddine-arbouch/Evo2-Variant-Intelligence` (frontend)
 - **Project category:** Genomic variant intelligence / Bioinformatics research tool
 - **Apparent project type:** SaaS / Research Tool / Academic Demo
 - **Public-facing or internal-facing:** Public-facing (deployed at <https://evo2-variant-intelligence.vercel.app>)
 - **Main user/customer:** Researchers, clinicians, and genomics professionals
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2. One-sentence factual description

A full-stack genomics research platform that integrates the Evo2 deep learning model for predicting genetic variant pathogenicity, combining a serverless Python API backend with a Next.js web interface that includes gene search, sequence visualization, ClinVar integration, and 3D protein structure viewing.

3. What the project does

- **Variant Pathogenicity Prediction:** Accepts genetic variants (SNVs, deletions, insertions) and returns a delta score predicting whether the variant is likely pathogenic or benign, calibrated on BRCA1 training data.
- **Multi-Modal Genomic Analysis Interface:** Provides an interactive console with five resizable panels for discovering genes, viewing gene context, visualizing 3D protein structures, exploring genomic sequences, and conducting variant analysis.
- **Session Management:** Users can create, save, and resume analysis sessions with persistent prediction history stored in a Supabase PostgreSQL database.
- **Real-time External Data Integration:** Fetches genomic sequences from UCSC Genome Browser API, gene metadata from NCBI, and clinical variant data from ClinVar, all with rate limiting and error handling.

- **3D Structure Visualization:** Integrates Molstar viewer to display PDB protein structures associated with genes being analyzed.
- **Prediction Caching:** Shares predictions globally across users for the same variant to avoid redundant API calls.

4. Confirmed implemented features

Feature	What it does	Evidence file paths	Confidence
Evo2 Model Integration	Loads and runs Evo2 7B deep learning model on H100 GPU via Modal serverless infrastructure; scores genomic sequences and calculates delta scores	<code>evo2-API/main.py:316-323</code> , <code>README.md:59-99</code>	High
Variant Analysis API	Accepts SNV, DELETION, INSERTION mutation types; validates input; fetches genome windows; constructs variant sequences; computes predictions	<code>evo2-API/main.py:328-472</code>	High
BRCA1-calibrated Classification	Uses empirically-determined threshold (-0.0009178519) and standard deviations from BRCA1 training data to classify variants as pathogenic/benign with confidence scores	<code>evo2-API/main.py:223-235</code>	High

Feature	What it does	Evidence file paths	Confidence
Gene Search (NCBI)	Searches NCBI gene database by symbol/name; returns 50 results with metadata; fetches genomic coordinates and summaries	<code>Evo2-Variant-Intelligence/lib/api.ts:342-510</code>	High
Sequence Fetching (UCSC)	Retrieves 8,192 bp genomic windows from UCSC API; validates coordinates; handles multiple genome assemblies (hg38, hg19, etc.)	<code>evo2-API/main.py:84-150</code> , <code>lib/api.ts:588-619</code>	High
ClinVar Integration	Queries NCBI ClinVar API for variants in genomic regions; returns clinical significance classifications	<code>lib/api.ts:691-745</code>	High
3D Protein Structure Viewer	Displays PDB structures via Molstar; maps genes to PDB IDs; allows manual structure selection	<code>Evo2-Variant-Intelligence/app/console/page.tsx:56-284</code>	High
Authentication (Supabase)	User session management with email-based login; protected console pages requiring authentication	<code>Evo2-Variant-Intelligence/app/console/page.tsx:68-93</code>	High
Session Persistence	Saves user sessions with gene selection and genome assembly choice;	<code>Evo2-Variant-Intelligence/app/console/page.tsx:95-150</code> , <code>supabase-migration.sql</code>	High

Feature	What it does	Evidence file paths	Confidence
	loads session state on revisit		
Prediction Caching	Global prediction cache shared across users for the same variant (position, chr, ref, alt) to avoid redundant inference	<code>Evo2-Variant-Intelligence/README.md:544-573</code>	High
Responsive UI with Resizable Panels	Discovery panel, gene context, sequence viewer, variant analysis, ClinVar, and structure panels with drag-to-resize capability	<code>app/console/page.tsx:232-346</code>	High
API Rate Limiting	Implements rate limiting for NCBI APIs (3 requests/second); exponential backoff retry logic for failed requests	<code>lib/api.ts:4-6</code> , <code>lib/api.ts:25-32</code> , <code>lib/api.ts:77-90</code>	High
Modal Serverless Deployment	Configured with H100 GPU, 3 max parallel containers, 120-second scaledown window, automatic retries	<code>evo2-API/main.py:297-303</code>	High
Vercel Frontend Deployment	Next.js 16 app optimized for Vercel; configured with environment variables for Supabase and Modal API	<code>Evo2-Variant-Intelligence/vercel.json</code>	High
Type Safety	Full TypeScript implementation on	<code>package.json</code> , <code>evo2-API/main.py:54-71</code>	High

Feature	What it does	Evidence file paths	Confidence
	frontend; Pydantic validation on backend; type-safe API contracts		

5. Partially implemented or unclear features

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Interactive Sequence Viewer Hover	Hover detection implemented with state management, but actual base-by-base visualization code not visible in excerpt; UI markup truncated	<code>app/console/page.tsx:165-173</code>	Verify sequence viewer renders individual bases and responds to hover
Complex Mutation Support	README mentions "complex mutations" in overview but only SNV, DELETION, INSERTION are implemented in the actual API	<code>README.md:37</code> , <code>evo2-API/main.py:57-61</code>	Confirm if DUPLICATION, INVERSION, TRANSLOCATION are truly supported or marketing language
Molstar 3D Interaction	3D viewer is embedded, but extent of user controls (zoom, rotate, select residues) not documented	<code>app/console/page.tsx:273-277</code>	Verify full Molstar feature set is exposed to users
Command Palette	Placeholder function exists but not fully implemented	<code>app/console/page.tsx:175-177</code>	Confirm if command palette is functional or planned future feature
Tour/Onboarding System	React Joyride integrated (<code>package.json</code> , <code>app/console/page.tsx:39, 226</code>), but tour content/steps not visible in code excerpts	<code>package.json:57</code>	Confirm tour steps are complete and tested

6. Mocked, hardcoded, or demo-only parts

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
RLS Policies (Development Mode)	Supabase migration SQL shows all-permissive RLS policies (<code>USING (true)</code>) with commented-out user-scoped policies for production	<code>supabase-migration.sql:55-82</code>	HIGH RISK: Publicly stating "secure user data isolation" when RLS is open; must be fixed before production use
Structure Mapping Fallback	If gene not found in structure map, UI shows placeholder "Select a structure from dropdown" message instead of real data	<code>app/console/page.tsx:280-282</code>	Medium: reasonable fallback, but clarify whether structure database is comprehensive
Package Name Placeholder	Frontend <code>package.json</code> uses placeholder name <code>"my-v0-project"</code> instead of production name	<code>package.json:2</code>	Low: cosmetic only, no functional impact
Agent Logging Instrumentation	Code contains inline agent logging calls to <code>http://127.0.0.1:7245/ingest/</code> (development debugging artifact)	<code>lib/api.ts:15-162</code>	MEDIUM RISK: Debug logging endpoint exposed in production code; should be removed before deployment

7. Technical stack

Layer	Technologies confirmed	Evidence file paths
Frontend Framework	Next.js 16 (App Router), React 19, TypeScript 5	<code>package.json:51, 53, 75</code>
Frontend UI	Tailwind CSS 4, Radix UI (18+ components), Lucide React icons, React Hook Form	<code>package.json:42, 13-39, 49, 56</code>
Frontend Components	React Resizable Panels, Molstar 3D viewer, Recharts, React Joyride	<code>package.json:58, 50, 59, 57, 60, 47</code>

Layer	Technologies confirmed	Evidence file paths
	tours, Sonner toasts, Embla Carousel	
Backend Framework	Python 3.12, FastAPI, Modal serverless platform	evo2-API/main.py:25, 14-15, 13, 47
GPU/ML Stack	NVIDIA CUDA 12.4, PyTorch, Transformer Engine, Flash Attention, Evo2 model (7B parameters, HuggingFace)	evo2-API/main.py:23-41 , README.md:59-67
Database	Supabase PostgreSQL, JSONB support for gene metadata	supabase-migration.sql:5-27
Authentication	Supabase Auth (JWT-based), Row Level Security (RLS)	Evo2-Variant-Intelligence/app/layout.tsx (implicit), supabase-migration.sql:51-82
External APIs	UCSC Genome Browser API (sequences), NCBI E-utilities (genes), NCBI ClinVar API (clinical variants), PDB (protein structures)	lib/api.ts:245-745
Deployment (Backend)	Modal (serverless GPU compute), persistent volumes for model caching	evo2-API/main.py:47-51
Deployment (Frontend)	Vercel (implied by deployment config and public homepage URL)	vercel.json , README.md:homepage
Package Manager	pnpm 9.12.1	package.json:78
Validation	Pydantic (backend), Zod (frontend)	evo2-API/main.py:17, 64-71 , package.json:64
State Management	React hooks (useState, useCallback), URL state via useSearchParams, server state via API	app/console/page.tsx:35-50
Testing	Not found in repo.	N/A
CI/CD	GitHub Actions workflows (health checks, pylint) referenced but files not examined	README.md:2-3

8. Routes, pages, and user flows

Route/Page	Purpose	Evidence file paths
<code>/</code> (homepage)	Landing page (content not examined)	<code>app/page.tsx</code> (exists)
<code>/signin</code>	Email-based login page	<code>app/signin/</code> directory, <code>app/console/page.tsx:75-76</code>
<code>/console?session={id}</code>	Main analysis interface; loads session state from URL parameter; requires authentication	<code>app/console/page.tsx:35-376</code>
<code>/dashboard</code>	Session management/list page (content not examined, but referenced as redirect)	<code>app/console/page.tsx:85</code>
<code>/docs</code>	Documentation/API reference page (content not examined)	<code>app/docs/</code> directory
POST <code>/api/analyze</code>	Proxy to Modal Evo2 API; accepts variant parameters; validates; returns prediction	<code>README.md:471-483</code>
GET <code>/api/clinvar</code>	Fetches ClinVar variants for genomic region; implements retry/rate limiting	<code>lib/api.ts:691-745</code>
GET/POST <code>/api/predictions</code>	Retrieve cached predictions by session; save new predictions to global cache	<code>README.md:512-573</code>
GET/POST/PATCH/DELETE <code>/api/sessions</code>	Full CRUD for user sessions; persistent storage in Supabase	<code>app/console/page.tsx:98-140</code> , <code>README.md:575-655</code>

9. Data model and backend logic

Database Schema (Supabase PostgreSQL)

Sessions Table:

- `id` (UUID, PK)
- `name` , `genome_assembly` , `selected_gene` (JSONB), `user_id` (FK to auth.users)
- Indexes on `user_id` and `updated_at`
- Auto-updating trigger for `updated_at`

Predictions Table:

- `id` (UUID, PK)

- `session_id` (FK), `position`, `chromosome`, `reference`, `alternative`, `delta_score`, `prediction`, `confidence`, `gene_symbol`
- Indexes on `session_id` and `created_at`
- Supports global cache sharing (same variant = same prediction across users)

API Endpoints

Backend (Modal):

- `POST /analyze_single_variant` - Variant analysis with authentication check

Frontend Proxy Routes:

- `POST /api/analyze` - Proxies to Modal, injects API key from env
- `GET /api/clinvar` - NCBI ClinVar search with retries
- `GET/POST /api/predictions` - Session prediction caching
- `GET/POST/PATCH/DELETE /api/sessions` - Session CRUD

Business Logic

Variant Scoring Algorithm:

1. Fetch 8,192 bp genomic window from UCSC API centered on variant position
2. Validate reference allele matches genome or auto-detect
3. Construct variant sequence (SNV substitution, deletion removal, insertion addition)
4. Score reference and variant sequences through Evo2 model (mean log-likelihood)
5. Calculate $\text{delta_score} = \text{variant_score} - \text{reference_score}$
6. Apply threshold classification (-0.0009178519 from BRCA1 data)
7. Calculate confidence as distance from threshold / standard deviation

Data Characteristics:

- **User data:** Real (sessions, predictions created by users)
- **Predictions:** Real inference results from Evo2 model (not cached mock data)
- **NCBI/UCSC/ClinVar data:** Real, live API calls to public databases
- **Gene metadata:** Real data fetched on-demand from NCBI with optional caching

10. AI-specific analysis

AI Model Used:

- **Provider:** HuggingFace (open-source)
- **Model:** Evo2 7B (7 billion parameters, StripedHyena 2 architecture)
- **Model URL:** `arcinstitute/evo2_7b`
- **Context Window:** 8,192 base pairs (optimized for genomic sequences)

AI Feature Implemented:

- Zero-shot variant effect prediction by comparing log-likelihood scores of reference and variant genomic sequences
- Single endpoint: scoring genomic sequences and computing delta score
- Threshold-based classification (not probabilistic neural network output)

Prompts:

- No prompts; model operates on raw genomic sequences (DNA sequences as text tokens)

Output Handling:

- Predictions are persisted to Supabase database (global cache)
- Inference results are real (not simulated); model runs on Modal GPU on every API call

Evidence of Real Inference:

- Detailed inference pipeline in code (sequence fetching, variant construction, model scoring, delta calculation)
- Integration with real UCSC genomic data
- Deployed on Modal with H100 GPU for computational requirements
- Calibration thresholds derived from actual BRCA1 analysis (referenced in code)
- Each variant generates unique delta score based on model inference

11. Portfolio table improvements

Field	Recommended Entry
Status	Prototype/Research Demo (public deployment at https://evo2-variant-intelligence.vercel.app , but with development-mode security policies and debug logging artifacts)
Industry	Bioinformatics, Genomic Variant Analysis, Personalized Medicine
Category	AI-Powered Research Tool, Genomic Intelligence Platform

Field	Recommended Entry
Project Type	Full-Stack SaaS Prototype with GPU-Accelerated ML Backend
What we built	A full-stack genomics research platform integrating Evo2 deep learning for variant pathogenicity prediction with real-time multi-API integration (UCSC, NCBI, ClinVar), interactive genomic visualization, 3D protein structure viewing, session management, and caching across a Next.js frontend and serverless Python backend.
Client/user problem	Researchers and clinicians need fast, accessible pathogenicity predictions for genetic variants to guide clinical decision-making and variant interpretation, combined with integrated access to gene context, protein structures, and clinical databases.
Main features	Evo2 AI variant pathogenicity prediction (SNV/deletion/insertion support), BRCA1-calibrated confidence scoring, gene discovery and metadata, UCSC sequence retrieval, ClinVar integration, 3D PDB structure visualization, session persistence, prediction caching, rate-limited API integration, responsive multi-panel UI.
Technical stack	Next.js 16 + React 19 + TypeScript (frontend), Python 3.12 + FastAPI (backend), Modal serverless GPU compute (H100), Evo2 7B AI model, Supabase PostgreSQL, NCBI/UCSC/ClinVar API integration, Molstar 3D viewer.
Proof available	Live public deployment, comprehensive technical documentation in README, architecture diagrams included, database schema, API reference, full source code with detailed comments.
Best sales angle	"Enterprise-grade genomic variant intelligence platform powered by foundation model AI, delivering real-time pathogenicity predictions with integrated clinical context and structure visualization for precision medicine workflows."
What should not be claimed	"Production-ready for clinical use" (database RLS policies are in development mode); "secure data isolation" (RLS not enforcing user scoping); "complex mutation support beyond SNV/deletion/insertion" (not implemented); "comprehensive feature completeness" (debug logging in production code, some features placeholder-level).

12. Safe public claims

✓ Confirmed safe to claim:

1. "Integrates the Evo2 7B deep learning model for genomic variant analysis"
2. "Predicts pathogenicity of single nucleotide variants, deletions, and insertions"
3. "Provides real-time UCSC genome sequence data integration"
4. "Displays ClinVar clinical variant data alongside predictions"

5. "Includes 3D protein structure visualization via PDB/Molstar"
 6. "Built with Next.js 16 + React 19 + TypeScript frontend"
 7. "Python FastAPI backend deployed on Modal serverless GPU infrastructure (H100)"
 8. "Supabase PostgreSQL database with session and prediction persistence"
 9. "User authentication and session management"
 10. "Interactive multi-panel genomic analysis interface with resizable components"
 11. "Gene discovery via NCBI API with metadata fetching"
 12. "Rate-limited NCBI API access with exponential backoff retry logic"
 13. "Global prediction cache to avoid redundant inference calls"
 14. "Deployed at <https://evo2-variant-intelligence.vercel.app>"
 15. "Open-source Evo2 model from HuggingFace"
 16. "BRCA1-calibrated classification thresholds"
 17. "API key authentication support"
 18. "Comprehensive API documentation and deployment guide"
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13. Unsafe or unproven claims

X Should NOT claim without evidence:

1. ~~"Enterprise-grade security"~~ — RLS policies are in development mode (all-permissive), not enforcing user data isolation
2. ~~"Production-ready for clinical use"~~ — Debug logging endpoints in production code; security policies incomplete; no audit trail
3. ~~"Complex mutation support (DUPLICATION, INVERSION, TRANSLOCATION)"~~ — Only SNV, DELETION, INSERTION are implemented; other mutation types in README but not in API
4. ~~"Real-time predictions"~~ — Accurate (model runs on GPU), but cold starts take 40-90 seconds
5. ~~"Fully-automated variant interpretation"~~ — Predictions are delta scores with thresholds, not comprehensive clinical interpretation
6. ~~"Clinical-grade variant database"~~ — Uses ClinVar (reliable), but application does not add proprietary clinical validation

7. ~~"Proprietary AI model"~~ — Uses open-source Evo2 model (from HuggingFace), not proprietary development
8. ~~"99.9% prediction accuracy"~~ — No accuracy metrics provided; only BRCA1 calibration data mentioned
9. ~~"Secure user data isolation"~~ — Row-level security policies are not enforcing user scoping in current state
10. ~~"Handles all genomic variant types"~~ — Limited to SNVs, deletions, insertions; no support for complex rearrangements

14. Missing evidence needed

To strengthen the portfolio case:

Evidence Type	What's Missing	Why It Matters
Screenshots	Console UI screenshots, session dashboard, 3D structure viewer in action	Demonstrates visual quality and user experience
Demo Video	2-3 minute walkthrough of gene discovery → variant analysis → ClinVar comparison → prediction	Shows workflow and interactivity
Performance Metrics	Prediction latency (warm/cold start), throughput (predictions/minute), accuracy vs. reference data	Proves scalability and reliability claims
Clinical Validation	Comparison of Evo2 predictions against ClinVar pathogenicity labels (sensitivity, specificity, AUC-ROC)	Validates AI model effectiveness for claimed use cases
User Testimonial	Quote from researcher or clinician using platform	Third-party credibility
Production Deployment Info	Current hosting cost, number of active users, monthly predictions run, uptime statistics	Proves real-world adoption and stability
Security Audit Report	Third-party penetration test or security review (especially for claimed "enterprise-grade")	Essential before claiming production readiness
Architecture Diagram	High-level visual of data flow (frontend → API → Modal → Evo2 → NCBI/UCSC → Supabase)	Clarifies system complexity
Case Study / Use Case	Real research project or clinical workflow enabled by platform	Demonstrates business value
Benchmark Comparison	How Evo2 predictions compare to other variant pathogenicity tools (e.g., SIFT,	Contextualizes model performance

Evidence Type	What's Missing	Why It Matters
	PolyPhen-2)	

15. Final confidence score

Confidence Score: 58/100

Reasoning:

✓ **Strengths (+):**

- **Technical implementation is solid:** Full-stack architecture is well-engineered with real AI model integration, proper API design, and comprehensive documentation
- **Real functionality:** Evo2 model runs on GPU, predictions are genuine (not mocked), external data integration is real
- **Production infrastructure in place:** Deployed on Vercel (frontend) and Modal (backend), uses professional services
- **Code quality:** TypeScript type safety, Pydantic validation, error handling, rate limiting, and retries all implemented
- **Reproducible:** Open-source model, publicly accessible code, clear deployment guide allows verification

✗ **Weaknesses (-):**

- **Security not production-ready:** RLS policies are all-permissive; this is a critical flaw for any data-sensitive application
- **Debug artifacts in production code:** Agent logging endpoints (127.0.0.1:7245) and #region markers should be removed before claiming production readiness
- **Feature claims exceed implementation:** Marketing language mentions "complex mutations" and "enterprise-grade" but evidence doesn't support these at the claimed level
- **No validation metrics:** Zero public proof of prediction accuracy, clinical relevance, or real-world performance; BRCA1 thresholds mentioned but no comparative benchmarks
- **Incomplete feature set:** Tour system, command palette, and some UI features are placeholders; documentation shows planned features that aren't fully implemented
- **Data privacy gaps:** Row-level security not enforcing user scoping in current state; unclear if production uses fixed version

What would increase score:

1. Remove debug logging and fix RLS policies (+15 points)
2. Publish clinical validation study or accuracy benchmarks (+12 points)
3. Share screenshots/video and use case examples (+10 points)
4. Security audit report or SOC 2 compliance evidence (+8 points)
5. Real production metrics (users, uptime, processing volume) (+5 points)

Current status: Suitable for **academic/research publication** or **portfolio demonstration of full-stack ML engineering skills**. **NOT suitable for claiming enterprise-grade clinical tool** until security, validation, and production-readiness gaps are addressed.